

Scheduling Cluster Tools in Semiconductor Manufacturing: Recent Advances and Challenges

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Abstract—Cluster tools are automated robotic manufacturing systems containing multiple computer-controlled process modules. They have been increasingly used for wafer fabrication. This paper reviews the modeling and scheduling methods for cluster tools with both nonrevisiting and revisiting processes. For nonrevisiting processes, we focus on the modeling and scheduling problems of cluster tools with different constraints. Then, their solution methods are reviewed and compared. For revisiting processes, this paper first discusses the scheduling problem of some general manufacturing systems with revisiting. Then, the modeling and scheduling methodologies used to solve the scheduling problems of cluster tools with revisiting processes are reviewed. Future research directions and conclusions are finally discussed.

Note to Practitioners—Semiconductor manufacturing systems are among the most advanced and complicated manufacturing systems. Their key equipment is highly automated robot-based cluster tools. With wafer residency time constraints, wafer revisiting, activity time variation, chamber cleaning requirements, and failure-prone process modules (PMs), it is very challenging to schedule and control them. This paper surveys their modeling and scheduling methods. Scheduling them requires one to schedule their robot tasks and processing activities simultaneously. Owing to wafer residency time constraints and the lack of buffers among PMs, it is difficult to conduct their optimal scheduling. This paper presents a thorough review of the state-of-the-art research results about modeling and optimal scheduling of clusters tools and indicates the future research directions.

Index Terms—Cluster tools, modeling and scheduling, semiconductor manufacturing, wafer fabrication.

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I. INTRODUCTION

MORE and more semiconductor manufacturers adopt cluster tools to process wafers to improve their chip production quality and productivity. Such tools provide a reconfigurable, flexible, and efficient environment [3], [11]. They integrate several computer-controlled process modules/chambers into a machine, each of which adopts single-wafer processing technology that handles wafers one by one at a chamber. As the wafer size increases and quality requirements become stricter due to circuit shrinkage, almost all fabrication processes adopt such an advanced technology [65].

Due to space restriction, a general radial-type cluster tool consists of no more than six process modules (PMs), an aligner module, a wafer handling robot (transfer module), and two loadlocks (LLs) for wafer cassette loading/unloading, which are mechanically linked together and computer-controlled. Each cluster tool has a wafer-handling robot that serves all PMs. The robot may have a single arm or dual arm, leading to a single-arm or dual-arm cluster tool. As shown in Fig. 1, PM, AL, and CL represent a PM, aligner, and cooler, and LA and LB represent LLs A and B, respectively. A multi-cluster tool is defined as a combination of several individual cluster tools that are interconnected through one or two-slot buffer modules [28], [81], [142]. In general, in scheduling a production system, one needs to schedule the job processing only [1], [2], [29], [118], while it needs to schedule both the robot and wafer processing concurrently for a cluster tool. Furthermore, in a cluster tool, it lacks buffer space and there are some process constraints. Thus, although a cluster tool looks simple, its scheduling and control are not trivial at all.

In a cluster tool, raw wafers enter the system through the LLs in a way of cassette by cassette. A cassette often holds 25 raw wafers with an identical recipe. After it is loaded into an LL, it is pumped into vacuum. Then, the wafers in it are ready for processing one after another. The robot unloads a wafer from the LLs and moves it to the aligner. A vision system at the aligner helps to adjust the wafer holding position on the robot arm. After an unloaded wafer is aligned, it is moved and loaded into a PM by the robot. After the completion of the processing at one or more PMs in a prespecified order, a wafer comes back to the LL where it comes from. After the last wafer in a cassette is unloaded from an LL,

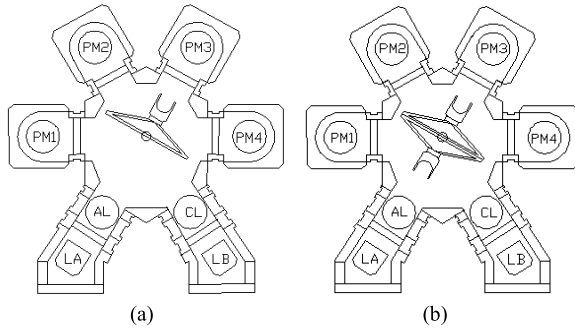


Fig. 1. Cluster tools with four PMs. (a) Single-arm robot. (b) Dual-arm robot.

processed and loaded back to it, the cluster tool starts processing wafers from the other LL, while the just finished cassette exits from the tool and a new cassette is prepared to be moved into it. In this way, wafer production continues.

A cluster tool is a piece of integrated equipment and requires high investment. It is thus important to operate it efficiently. Seeking efficient and effective scheduling and control methods for cluster tools represents an important research area in semiconductor manufacturing industry. Especially, many practical issues make the operation of a cluster tool very challenging.

Some wafer fabrication processes, such as low-pressure chemical vapor deposition (CVD), impose a strict constraint on the wafer sojourn time in a PM, which requires that a completed wafer should be removed from a PM within a given short time. This requirement is called wafer residency time constraints, which make the scheduling problem of cluster tools very complicated.

In operating a cluster tool, there are random disturbance to activity time. For example, unpredictable time variation arises from wafer alignment failure, PM processing delay, communication delay, and computer processing delay [58]. The activity time variation greatly complicates the operation of cluster tools, since it can result in the violation of wafer residency time constraints, thereby leading to wafer quality problems.

In a cluster tool, PMs are failure prone [141], though the failure rate for a specific PM is usually small. A failure can reduce productivity and scrap a wafer if it is not properly handled. With a large number of cluster tools in a fab, it is important and challenging to deal with such failures.

The residual chemicals and impurities in a chamber can contaminate the chamber, such that a wafer may be damaged. Thus, a chamber needs to be cleaned during the operation when significant contamination is detected [50].

With mass customization, the lot size becomes smaller and smaller. Thus, a tool needs to frequently switch from processing one lot to another such that there is a transient process. It is also a challenge to operate a tool under such a transient processes.

With the above challenging issues for operating cluster tools, it is crucial to develop efficient and effective scheduling techniques for them. Great attention has been paid to do so in the past two decades. About eight years ago, Lee [65]

presented a review of scheduling methods of cluster tools in semiconductor manufacturing but mentioned no future work in any detail. Since then, there are newly created and yet important research demands from semiconductor manufacturing industry and new development of scheduling and control methods to meet them. For example, manufacturers have to increase the wafer size from 200 to 300 mm even to 450 mm. They have to reduce the semiconductor circuit width such that a wafer can contain more chips, thus leading to lot size reduction [53]. When the consecutive lots hold different-type wafers, a cluster tool often needs to switch from processing one type of wafers to another, such that there are more transient processes in wafer fabrication. Therefore, it is urgent to develop efficient approaches to schedule such transient processes for cluster tools. The new and urgent demands for seeking efficient and effective scheduling and control methods for cluster tools motivate us to perform this survey.

The remainder of this paper is organized as follows. Section II provides the wafer flows and scheduling strategies. Section III reviews modeling and scheduling methods for cluster tools with nonrevisiting processes. Section IV discusses the existing methods for modeling, scheduling, and control of cluster tools dealing with revisiting processes. Section V presents the future research directions. Section VI draws a conclusion.

II. WAFER FLOWS AND SCHEDULING STRATEGY

A. Wafer Flow Patterns

A cluster tool is reconfigurable, since a PM in a tool can be configured to complete different fabrication processes. Multiple PMs may be configured for the same fabrication process and they are called parallel PMs for the process. A fabrication process has a unique working environment and is defined as a fabrication step. There are a number of operations for processing a wafer. If every operation of a wafer needs to be completed at different steps, we have a nonrevisiting process, while if two or more operations of a wafer need to be completed at the same step more than once, we have a revisiting process.

Different types of wafers have different recipes, leading to different processing routes. For a cluster tool without revisiting, its wafer flows are similar to conventional flow lines or assembly lines. However, different from the latter, there is no intermediate buffer between PMs and the wafer delivery between PMs must be performed by the robot equipped for a cluster tool. The wafer flow pattern for a nonrevisiting process can be denoted as $(m_1, m_2, \dots, m_n)$, where m_i gives the number of parallel PMs at Step i and n is the number of total fabrication steps. With no wafer revisiting, each operation is processed at different steps. Fig. 2(a) shows a single-arm tool with wafer flow pattern $(1, 1, 1, 1)$, implying that each step has only one PM. In Fig. 2(a), the tool has four PMs denoted as PM₁–PM₄ that are configured for Steps 1–4, respectively. In Fig. 2(b), it shows a single-arm tool with wafer flow pattern $(1, 2, 1)$, where PM₁ and PM₄ are configured for Steps 1 and 3, respectively, while both PM₂ and PM₃ are for Step 2.

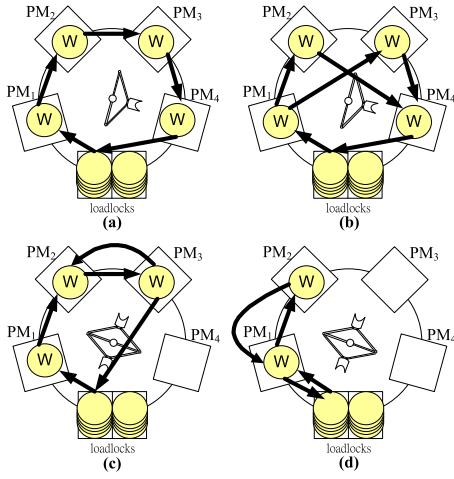


Fig. 2. Different wafer flow patterns. (a) Single-arm tool with wafer flow pattern (1, 1, 1, 1). (b) Single-arm tool with wafer flow pattern (1, 2, 1). (c) Dual-arm tool with wafer flow pattern (1, (1, 1)^h). (d) Dual-arm tool with wafer flow pattern ((1)₂⁰, 1).

A cluster tool with a revisiting process is no longer a flow shop. Its wafer flow pattern can be denoted as $(m_1, \dots, (m_i)_f^k, \dots, (m_g, \dots, m_j)^h, \dots, m_n)$, where $n > j > g > i \geq 1$. $(m_i)_f^k$ denotes that, after executing Step f , Step i is repeated once, and then the wafer is delivered to Step k for the next operation. Note that if it requires repeating a single step, it repeats only once. $(m_g, \dots, m_j)^h$, $h \geq 2$, denotes that, after being processed at Step $g - 1$, a wafer is processed at Steps g to j sequentially. After the processing at Step j , the wafer is processed at Steps g to j sequentially again. This process is repeated for $h - 1$ times. When the revisiting process is completed, the wafer is delivered to Step $j + 1$ by the robot. Let Step 0 denote the LLs. Fig. 2(c) shows an example with wafer flow pattern (1, (1, 1)^h), where PM₁–PM₃ are configured for Steps 1–3, respectively. Steps 2 and 3 together form a revisiting process. Fig. 2(d) shows an example for a process with wafer flow pattern ((1)₂⁰, 1). In this example, PM₁ and PM₂ are configured for Steps 1 and 2, respectively. A wafer is processed at Steps 1 and 2 sequentially. Then, it is moved back to Step 1 for being processed. After it is processed at Step 1 for the second time, it is returned to the LLs.

Cluster tools in semiconductor manufacturing mainly operate under a nonrevisiting process. Thus, this paper mainly reviews the research advancements in such a process; while briefly introducing some results for the revisiting processes.

B. Backward and Swap Strategies

In practice, the robot moving time among the PMs can be treated as a constant and is much shorter than the wafer processing time [58]. Thus, a backward strategy is optimal for single-arm cluster tools [72], [81], while a swap strategy is shown to be efficient for dual-arm cluster tools [114].

For a single-arm cluster tool, with a backward strategy, the robot performs the following activities in a cycle: unloads a completed wafer from Step n → delivers it to the LLs → moves to Step $n - 1$ → unloads a wafer from Step $n - 1$ → delivers it to Step n → ... → moves to Step i →

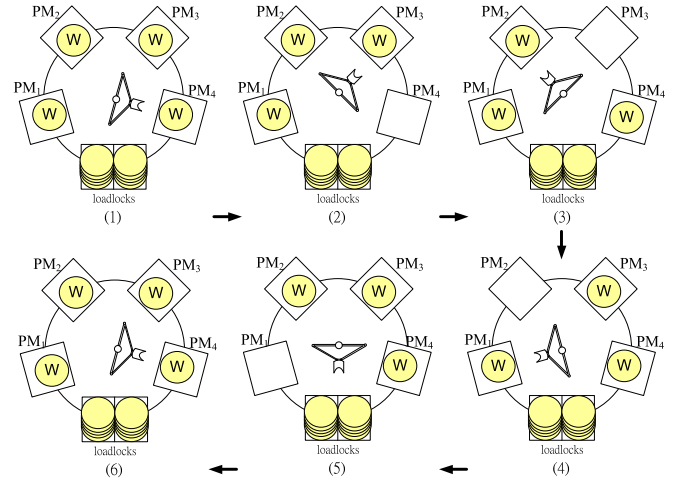


Fig. 3. Backward strategy.

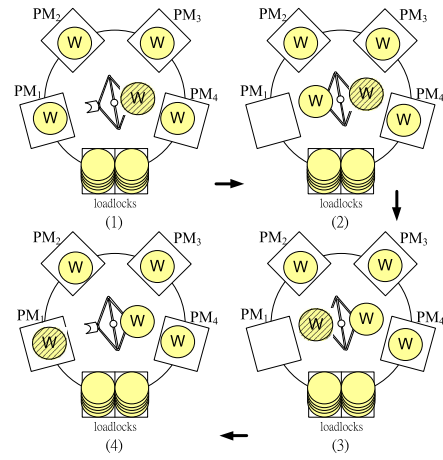


Fig. 4. Swap operation at Step 1.

unloads a wafer from Step i → delivers it to Step $i + 1$ → moves to Step $i - 1$ → unloads a wafer from $i - 1$ → delivers it to i → ... → moves to Step 1 → unloads a wafer from Step 1 → delivers it to Step 2 → moves to the LLs → unloads a raw wafer from the LLs → delivers it to Step 1 → moves to Step n . Fig. 3 shows an example with wafer flow pattern (1, 1, 1, 1) by using a backward strategy.

For a dual-arm cluster tool, with a swap strategy, the robot performs the following activities in a cycle: unloads a wafer from the LLs by one of arms → moves to Step 1 → unloads a wafer from Step 1 by the other arm → rotates → loads a wafer into Step 1 → ... → moves to Step i → unloads a wafer from Step i by the empty arm → rotates → loads a wafer into Step i → ... → moves to the LLs → loads a completed wafer into the LLs. Fig. 4 shows a swap operation at Step 1 for a dual-arm cluster tool with wafer flow pattern (1, 1, 1, 1).

For a single-arm cluster tool, under the steady state with a wafer being in each PM, the system is deadlock-prone if a backward strategy is applied. For example, assume that the system is at the state shown in Fig. 3(1). At this state, if the robot moves to PM _{i} , $i \in \{1, 2, 3\}$, or the LLs instead of PM₄ and unloads a wafer there, and then it moves to the next step for loading the wafer into the PM. Then, the system is deadlocked, since the PM is occupied. With a backward

strategy, in a cycle, the robot visits the steps sequentially as: Step $n \rightarrow$ Step $n-1 \rightarrow \dots \rightarrow$ Step 1. For a dual-arm cluster tool, the robot has two arms named Arms 1 and 2, respectively. With Arm 1 holding an unprocessed wafer and Arm 2 being empty, to perform a swap operation at a PM, Arm 2 can be treated as a temporary buffer and used to unload a wafer from the PM such that Arm 1 can load the unprocessed wafer into the PM. In this way, a swap operation is finished. Thus, with a swap strategy, under the steady state, in a cycle, the robot visits the steps sequentially as: Step 1 $\rightarrow$ Step 2 $\rightarrow \dots \rightarrow$ Step n . Notice that this sequence is totally different from a backward strategy for a single-arm tool. In addition, the robot tasks at each step for single-arm and dual-arm cluster tools are different, which makes their modeling and scheduling different.

III. NONREVISITING PROCESSES OF CLUSTER TOOLS

A. Manufacturing System Scheduling

To realize full automation and increase efficiency, manufacturing cells are usually served by a single material handling device, such as a robot, an automated guided vehicle, or a hoist for loading, unloading, and moving parts among machines. When such a device is a robot, the corresponding systems are often called robotic cells. Cluster tools are such a kind of cells. The important issues to be considered are cell design, robot movement, and part processing sequence, to name just a few.

Extensive studies have been done for robotic cells [10], [25]–[27], [30], [37], [38], [108], [109], [111]. Under the assumption that there is no restriction on how long a completed part can stay at a machine, these studies mainly focus on robotic flow shop with a single machine for each step. However, cluster tools in semiconductor fabrication exhibit some differences not seen in traditional robotic flow-shop systems.

- 1) There is no intermediate buffer between PMs.
- 2) For some processes, strict wafer residency time constraints are imposed, i.e., a wafer cannot stay in a processing chamber for too long after it is done.
- 3) A processing step can have parallel PMs to perform the same processing function.

With no intermediate buffer in a cluster tool, one needs to schedule wafer processing and robot activities simultaneously. At the same time, an obtained schedule must meet wafer residency time constraints if any. All these differences make the scheduling problem of cluster tools more complicated and difficult than the one for a traditional robotic flow shop.

The hoist system is somehow similar to a robotic flow shop except that there is a part processing window constraint [14]–[16], [48], [79], [82]. However, a hoist system differs from a cluster tool in the following aspects.

- 1) In the former, there is only one processing unit for each step, while in the latter, more than one PM can be configured for a step.
- 2) With a radial layout, the movement of the robot in a cluster tool is much more efficient than a hoist.
- 3) A hoist can hold one part at a time, while the robot in a dual-arm cluster tool can hold two.

Hence, different techniques for modeling, analysis, scheduling, and control of cluster tools are required.

A cluster tool is a flow-shop system when there is no wafer revisiting. There are many studies on scheduling flow-shop systems [1], [41], [80], [83], [89]–[91]. These studies focus on minimizing makespan with limited buffer size. When different types of wafers with different recipes are processed in a cluster tool simultaneously, such a tool is similar to hybrid shops. In [6], [8], [9], [29], and [40], hybrid shop scheduling problems are studied. In these studies, heuristic algorithms are mainly presented, such as hybrid PSO-based algorithms, beam search methods, tabu search-based algorithms, and genetic algorithms. However, in cluster tools, one needs to deal with some issues, such as wafer residency time constraints, chamber cleaning requirements, and single wafer transportation module. Consequently, the obtained results in the above studies are not applicable to cluster tools.

B. Scheduling Without Wafer Residency Time Constraints

With two LLs, when wafers in one cassette in one LL are being processed, the other LL can be used for loading/unloading another cassette of wafers. In this way, a cluster tool can continue its operations without being interrupted, i.e., it operates in a steady state for most of time. Great efforts have been done in the modeling and performance evaluation of cluster tools under the steady state [28], [87], [110], [114], [144], [155]. It is found that the operations of cluster tools can be divided into two regions: process bound and transport bound. In the former, the robot has idle time and wafer processing time in PMs determines the system cycle time. In the latter, the robot is always busy and the total robot task time in a cycle determines the cycle time. Note that most cluster tool operations in wafer fabrication belong to the former and thus the former is the focus of most cluster tool scheduling studies.

Operating a cluster tool needs to schedule its robot and PM processing activities simultaneously. However, PM activities follow the robot tasks, since a PM can start its wafer processing only after a wafer is loaded into it [107]. Hence, it is most critical to schedule the robot tasks well. Robot activities fall into two types: 1) unloading a wafer from and loading into an LL, moving, swapping, and loading/unloading at a PM and 2) waiting. A wafer should be loaded into or unloaded from a concerned PM at the right time. Such a time is dependent on which steps and how long the robot waits. Thus, it is very important to determine if the robot should wait before or after performing a task for a process-bound cluster tool. In other words, to schedule such systems well, one needs to determine the robot activity sequence and robot waiting time.

By using timing diagrams, Perkinson *et al.* [87] analyze the relationship among processing time, transport time, and maximum throughput of a single-arm cluster tool. Based on such diagrams, Venkatesh *et al.* [114] analyze the throughput of dual-arm cluster tools. Yim and Lee [144] formulate the scheduling problem by using the job requirement table with random keys. Through the negotiation between the operations in the candidate list and the current state of a system, a deadlock-free schedule is derived.

Petri nets (PNs) are widely used to model discrete-event systems for analysis, simulation, and control [18]–[22], [24], [33], [42], [43], [74]–[77], [112], [134], [146]–[150], since they can model the concurrent activities well. These models belong to process-oriented PNs (POPNS). In POPNS, operation places are used to describe the operations that every part has to go through, and resource places are used to describe the resource requirements for all the operations. Thus, the manufacturing processes are clearly described. However, the size of a POPN model increases rapidly as the number of processes increases. To avoid this problem, Wu [116], [117] and Wu and Zhou [122] first proposed resource-oriented PNs (ROPNS) to model such systems. They view a part manufacturing process as a part visiting the resources one by one according to its prescribed processing routes, and colors are introduced to describe different part flows. In ROPNS, each resource is modeled by a single place and the part manufacturing processes are modeled by the paths along which the parts visit the resources. Since the part manufacturing processes are modeled in an implicit way, the model is very compact, its size is linear with the number of resource types, and its useful structural characteristics can be easily revealed. The successful applications of ROPNS for modeling, control, and scheduling can be found in [119], [123]–[125], [129], and [131].

Zuberek [155] uses timed marked graph (a subclass of POPNS) to model the steady state of both single-arm and dual-arm cluster tools. The system cycle time is calculated as follows: 1) find all loops in the model; 2) divide the sum of all transition time delays over the total number of tokens in every loop to obtain the cycle time for every loop; and 3) find the maximum loop cycle time as the system cycle time.

An individual cluster tool involves only one robot. In a multicluster tool, multiple robots run independently and co-operatively. Some buffers among individual tools are not only pure buffers but carry some processing functions. Therefore, a multicluster tool is more complicated to control and schedule than an individual cluster one. It may be decoupled into individual cluster tools to analyze its operations. Based on the divide-and-conquer method, Yi *et al.* [142], [143] present a decomposition method to analyze the steady-state throughput given a robot schedule, which calculate the fundamental period (FP) of individual cluster tools and then uses $\max(FP_1, FP_2, \dots, FP_n)$ as the system FP. Note that the time between subsequently completed wafers arriving at the LL is the FP.

Chan *et al.* [12], [13] decouple multicluster tools and propose a polynomial algorithm for a two-cluster tool and an exponential algorithm for a k -cluster tool ($k > 2$) to calculate its minimal cycle time. This is a significant advancement in scheduling multicluster tools. In fact, this method can be used to obtain an optimal solution in the sense of multiwafer cyclic scheduling. However, it gives rise to a question whether there is always a one-wafer cyclic (1-WC) schedule for a k -cluster tool composed of k individual cluster tools such that the lower bound of cycle time is reached? To tackle this issue, Zhu *et al.* [151] propose a method to carefully assign robots' waiting time and obtain an optimal 1-WC schedule with the system being modeled by a colored-ROPN. They for the first

time establish the easy-to-check conditions under which the optimal schedule can reach the lower bound of the cycle time for multicluster tools. Then, based on their work, further studies are conducted for multicluster tools in [4], [137]–[140], [152], and [153].

In a typical large-scale fab, over 500 overhead hoist transfers (OHTs) are used to deliver wafer cassettes among about a 1000 of cluster tools [52]. With so many OHTs and cluster tools, it is very complicated to dispatch wafer lots to tools by OHTs so as to minimize the idle time of tools and avoid OHT-related deadlock. Therefore, recently, in semiconductor fabs, cluster tools are arranged in a line with a buffer to form a new system called in-line multiple cluster tools, which are different from the traditional multiple cluster tools in [142], [143], and [151]. Kim *et al.* [52] develop mixed integer programming models to minimize the makespan of producing 25 wafers for in-line multiple cluster tools. They also propose multimode operations. In other words, if the workloads of wafers in a component tool are small, multiple wafer types are processed concurrently in such a tool, otherwise only one wafer type is processed in it. In this way, the utilization rate of this component tool is improved.

The determination of FP of multicluster tools is similar to the approaches to determine the cycle time in an assembly line balancing problem. In [5], [32], and [86], studies are conducted to minimize the cycle time in an assembly line balancing problem. However, their results are not applicable to cluster tools since: 1) in a multicluster tool, it is important to coordinate the robots of individual cluster tools in an efficient way; 2) for cluster tools, there are wafer residency time constraints for some processes; and 3) there is no buffer among PMs in individual cluster tools.

We conclude that, given single-cluster tools without wafer residency time constraints, the techniques for their deadlock control and scheduling under the steady state are mature based on the developed models.

C. Scheduling With Wafer Residency Time Constraints

The studies in [12], [13], [28], [87], [110], [114], [142], [143], [151], and [155] are conducted under the assumption that a wafer can stay in a PM for unlimited time. Some processes, for example, low pressure CVD, have strict wafer residency time constraints. When a wafer completes its processing, it should be unloaded from a PM within a rather limited time; otherwise, the wafer surface quality deteriorates due to residual heat and gas within the PM, and the entire wafer may be scrapped.

Some traditional manufacturing processes require that after an operation is completed, the next operation should be started within a limited time. Wafer residency time constraints are similar to such a requirement. The difference is that, in traditional processes, one does not need to schedule a part delivering process, but wafer fabrication in cluster tools requires to do so. Branch-and-bound (B&B) algorithms are developed to schedule a two-machine flow shop with such a requirement in [2], [7], [46], and [136]. These methods are computationally time-consuming.

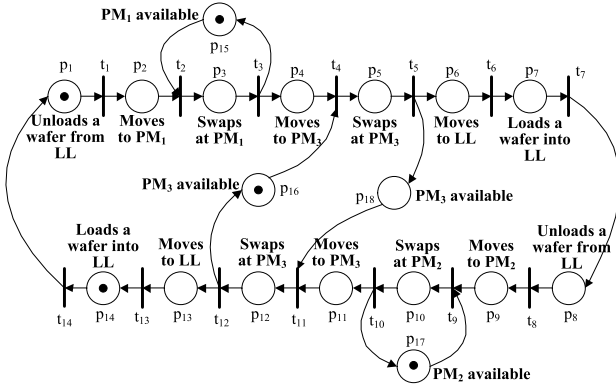


Fig. 5. Timed marked graph for wafer flow pattern (2, 1) for a 3-PM tool.

For cluster tools, it is complicated to find a feasible schedule to meet wafer residency time constraints. To do so, for a dual-arm cluster tool, Rostami *et al.* [106] give a search method combined with heuristics. For some cases, there are residency time constraints on robots, i.e., robot residency time constraints. In such cases, a wafer can reside on robots within limited time, and otherwise it is damaged by the air. Then, the methods in [106] are further extended to schedule dual-arm cluster tools with both wafer and robot residency time constraints, and intermediate buffers [104], [105]. Although this method can find an optimal solution, it is neither a systematic method nor computationally efficient.

Kim *et al.* [58] propose a systematic scheduling method for dual-arm cluster tools with wafer residency time constraints. They model a tool by timed event graphs or timed marked graphs (a subclass of POPNs), as shown in Fig. 5 with wafer flow pattern (2, 1), where PM₁ and PM₂ are configured for Step 1 and PM₃ for Step 2, and places p_1 – p_{14} denote the robot tasks and p_{15} – p_{18} denote three PMs. When a token is in p_i , $i \in \{15, 16, 17, 18\}$, p_i is available. Note that the transitions are just used to connect the places. Based on this model, the scheduling problem is formulated as a linear programming problem. For the first time, they study the schedulability issue in scheduling such cluster tools. However, their obtained schedulability condition is sufficient but not necessary.

The model is extended by using negative places and tokens in [73]. They derive a necessary and sufficient schedulability condition under which the system is schedulable. If schedulable, an optimal schedule can be found by finding all the circuits in the model, which may not be computationally efficient since theoretically the number of circuits in the model grows exponentially with the model size. Notice that a model in [58] and [73] is not generic, since it is wafer flow pattern-dependent. Thus, for different wafer types, different models are needed, which limit the application of their proposed methods. By using the similar modeling methods, Jung *et al.* [47] propose a B&B algorithm for cyclic scheduling of cluster tools and robotic flow shops.

Yoon and Lee [145] propose a scheduling algorithm with two subalgorithms, i.e., FEASIBLE_SCHED_SPACE and OPTIMAL_SCHED. The former computes the feasible solution space in a continuous time domain, and the latter

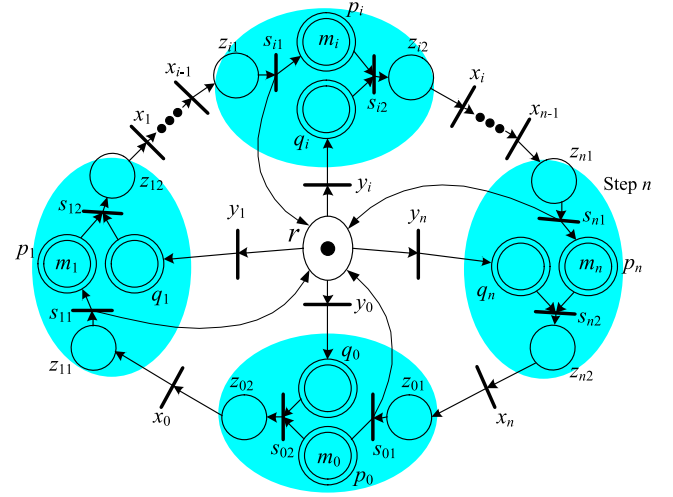


Fig. 6. ROPN model of a single-arm cluster tool.

finds the optimal solution that minimizes the completion time of the last operation for a newly inserted wafer. This online scheduling algorithm is of polynomial complexity. However, it is applicable to noncyclic released wafers only, i.e., multiple wafer types are produced in cluster tools and wafer types have different processing routes.

To achieve high computational efficiency, Wu *et al.* [119] propose an ROPN-based scheduling analysis of single-arm cluster tools with wafer residency time constraints. They present an ROPN model for the system, as shown in Fig. 6. In this model, the LLs are treated as a step (Step 0), and place p_i is used to model Step i with capacity m_i indicating that there are m_i PMs. Place r is used to model the single-arm robot. Transitions s_{i1} and s_{i2} model the loading and unloading of a wafer into and from a PM at Step i , respectively. Transitions x_i and y_i model the robot moving between two steps with and without carrying a wafer, respectively. Place z_{ij} is just used to connect transitions. Different from the PN model in [58], the robot waiting is modeled as an event and denoted by place q_i in the ROPN model, as shown in Fig. 6. In this model, time is associated with both transitions and places.

By the ROPN model in [119], the robot waiting is explicitly described such that the relationship between wafer sojourn time at a PM and the system cycle time can be well revealed. In this way, it is found that the key to schedule a cluster tool with wafer residency time constraints is to regulate the robot waiting time. With this perspective, based on the model, the scheduling and control problem of cluster tools are studied from a control-theoretic perspective instead of mathematical programming formulations. Then, necessary and sufficient schedulability conditions are derived in [119]. If schedulable, an algorithm that is given by simple analytical expressions is proposed to find a feasible and optimal schedule. The results obtained in [119] are then extended to the scheduling problem of dual-arm cluster tools with wafer residency time constraints [125].

It follows from the above discussion that, under the assumption that the activity time is deterministic and known, the periodic scheduling problem of single-cluster tools is well solved.

D. Scheduling With Activity Time Variation

In the above discussions, with residency time constraints or not, the robot activity time and wafer processing time are assumed to be deterministic. In fact, in operating a cluster tool, some abnormal events can occur. For example, unpredictable variation arises from wafer alignment failure and retrieval with failure probability being 3%–10% depending on cluster tools, PM processing delay, communication delay, and computer processing delay. Hence, activity time is subject to random variation and sometimes abrupt random disturbance [58].

With wafer residency time constraints, activity time variation may make a feasible schedule obtained under the assumption of deterministic activity time infeasible. It is necessary to find an efficient scheduling and control method such that a tool is robust to the random disturbance on the task time. Aiming at this issue, Kim and Lee [51] address the schedulability analysis issue by confining time variation within finite intervals for the first time. Based on the earliest start strategy, they identify so-called always schedulable and never schedulable cases. By adopting a branching technique, a schedule can be found such that the residency time constraints can be satisfied if it is always schedulable. If not, they leave it as an open problem.

Wu and Zhou [126]–[128] find that the robot waiting time can be dynamically adjusted to offset the effect of activity time variation on the wafer sojourn time fluctuation in a PM. To do so, an ROPN model is developed to describe the behavior of the system. With it, a real-time control policy is proposed to dynamically adjust the robot waiting time. The basic idea is as follows. Assume that, by an off-line schedule, at Step i , the robot is scheduled to wait for ζ_i time units. In the real-time, if it is observed that the robot arrives at Step i ν_i time units later. Then, if $\nu_i \leq \zeta_i$, the robot waiting time at Step i is shortened to $\zeta_i - \nu_i$; otherwise if $\nu_i > \zeta_i$, it waits no time. In this way, the wafer sojourn time fluctuation caused by activity time variation in a PM can be offset as much as possible. Based on this idea, a real-time control policy is developed to do so. By this policy, it is shown that some never schedulable cases identified in [51] are in fact schedulable.

Then, based on the real-time control policy, an off-line schedule is designed by determining the robot waiting time. The off-line schedule together with the real-time control policy forms a real-time schedule. Such a schedule is shown to be optimal in terms of productivity. Notice that, the real-time control policy is formed by simple control laws given analytically. Therefore, their proposed method is very efficient.

For dual-arm cluster tools, only the robot waiting time during swapping affects the workload at a processing step and all the others have no effect on the workload. However, for single-arm cluster tools, any scheduled robot waiting time change has its contribution to the workload. Hence, scheduling single-arm cluster tools with activity time variation is much more complicated according to [92] and [93]. Based on the studies in [126]–[128], efficient methods for scheduling single-arm cluster tools with residency time constraints and bounded activity time variation are proposed in [85], [92], and [93].

With the randomness of the activity time variation and real-time requirement for operating a cluster tool in this case,

it is extremely difficult to perform its scheduling and control. It follows from the above discussion, there are many issues to be addressed despite much progress has been made in handling activity time variation.

E. Transient Process Scheduling

While most aforementioned studies on cluster tool scheduling are conducted for finding an optimal periodic schedule, some studies have been done on transient process scheduling [49], [53]–[57], [66]–[68], [100], [115], [154] due to their increasing importance. Given a robot task sequence, a transient period for the start-up and close-down processes is minimized in [49] for a dual-arm cluster tool. For single-arm cluster tools, scheduling methods are proposed for noncyclic scheduling problem in [55] and [115]. With small batches, lot switching occurs frequently. Studies are conducted and techniques are developed for scheduling lot switching processes for both single and dual-arm cluster tools [53], [56], [57], [66]–[68]. With wafer residency time constraints, linear programming models are developed to find optimal schedules for start-up and close-down processes of single-arm cluster tools in [100] and [154]. Furthermore, with the assumption that a robot task sequence is given, Kim *et al.* [54] propose the conditions under which there always or never exists a feasible schedule for noncyclic operation of time-constrained cluster tools with time variation. They also develop conditions under which a given schedule that is neither always schedulable nor never schedulable can become always schedulable by adding a control arc into the developed PN model.

In a cluster tool, PMs are failure prone [141]. Although a tool must enter a close-down process to reach a full stop if a nonparallel PM fails, the system may be able to operate in a performance degraded way when a parallel module fails [58], [107]. In fact, before a cluster tool enters the “new” steady state after failure from the one before failure, it should go through a transient process. Thus, in this case, it needs to schedule the transient process caused by PM failures.

Since the transient process caused by PM failures is different from that of start-up, close-down, and lot switching, it is important to schedule this transient process such that the wafer residency time constraints are met. Although efficient methods have been proposed for transient process scheduling in [49], [53]–[57], [66]–[68], [100], [115], and [154], they are not applicable to the scheduling problem of cluster tools with failure-prone PM, since no approaches are given to deal with such PM failures.

With wafer residency time constraints, any failure of PMs may make the existing schedule infeasible. Thus, it is crucial to respond to a PM failure such that the system can keep working in a feasible way if possible. To do so, for single-arm cluster tools, based on ROPN models, Qiao *et al.* [95], [97] propose response policies to the PM failure such that the tools can successfully undergo a transient process from the steady state before failure to the “new” one after failure without violating wafer residency time constraints.

The above results represent the progress for dealing with transient process scheduling of cluster tools. However, more research is necessary on this issue to be discussed later.

F. Comparison of Modeling and Scheduling Methods

Based on the above discussion, Table I in the Appendix summarizes the results for modeling cluster tools dealing with nonrevisiting processes in terms of: 1) tool types, i.e., single-arm and dual-arm tools; 2) modeling methods; 3) the states that a model can describe; 4) advantages of a model; and 5) disadvantages of a model. Table II in the Appendix summarizes the results for scheduling cluster tools dealing with nonrevisiting processes in terms of: 1) tool types, i.e., single-arm, dual-arm, and multicluster tools; 2) scheduling methods; 3) advantages of a scheduling method; 4) disadvantages of a scheduling method; and 5) objectives.

IV. WAFER REVISITING PROCESSES

A. Manufacturing Systems With Revisiting

Manufacturing systems have mainly been treated as either job shops or flow shops. For the former, parts may arrive with random routes, while, for the latter, the routes are fixed and acyclic, such as assembly lines [59], [150].

For flow shops with revisiting, heuristic algorithms are developed to minimize makespan in [17] and [45]. In [31], studies are conducted on multiobjective resolution of a reentrant hybrid flow-shop scheduling problem. To solve such a problem, an efficient multiobjective method is proposed to maximize the utilization rate of the bottleneck and minimize the maximum completion time in [31]. Gomes *et al.* [39] present a mathematical programming approach to a make-to-order flexible job shop with a reentrant process and assembly.

Kumar [60] develops a queueing network model for a reentrant line in semiconductor manufacturing to analyze the performance by using a simulation technique. With a queueing network model suitable for a wafer fabrication reentrant line, two scheduling policies for system stability are discussed in [61]. A two-layer hierarchical control method is proposed for the schedule optimization problem of reentrant semiconductor manufacturing lines in [113]. Jeng *et al.* [44] show that a semiconductor manufacturing system belongs to a class of large and complex discrete-event systems. They point out that the complexity arises from a large number of re-entrant production procedures and shared resources as well as the complicated interactions between procedures and resources. Thus, it is the reentrance and a variety of constraints that make a semiconductor manufacturing system more difficult to schedule than a classical flow shop or job shop.

The research results obtained for scheduling reentrant flow shop and semiconductor manufacturing systems are not applicable for scheduling cluster tools since: 1) parallel PMs may exist in a cluster tool; 2) there is no buffer space among its PMs; 3) it has only one transport module (one robot with single arm or dual arms); and 4) it must meet residency time constraints for some processes.

B. Modeling for Cluster Tools With Revisiting

Although most of wafer fabrication processes in a cluster tool are a nonrevisiting one, there are research reports on scheduling cluster tools with wafer revisiting. We briefly review the existing results as follows.

In wafer fabrication, there are a number of processing steps and each step has a unique working environment. If a process requires a wafer to visit a step only once, it is a nonrevisiting process, and otherwise it is a revisiting one. With wafer revisiting, a cluster tool is no longer a flow shop, therefore invalidating all the existing scheduling methods developed for processes without wafer revisiting. Furthermore, sharing a PM by multiple operations greatly complicates its scheduling problem and even causes deadlocks according to [71] and [120]. It is known that it is very difficult to optimally schedule a deadlock-prone system.

Digraphs, automata, and PNs are three major mathematical tools to handle deadlock problems in discrete-event systems [36]. Graph theory or a digraph is simple and intuitive tool to describe interactions between operations and resources. Based on it, a deadlock-control policy can be developed [23], [34], [35], [62], [132], [133]. Also, many effective and efficient deadlock-control policies are developed based on automata [63], [64], [102], [103], [135]. Due to their ability to deal with concurrent activities, PNs are widely used as a modeling and analysis tool for flexible manufacturing systems [117], [122]–[124], [146], [148], semiconductor manufacturing [78], [149], and cluster tools [58], [155].

For cluster tools with chamber revisiting, Zuberek [156] shows that chamber revisiting can be systematically described by a set of discrete configurations that can also be used to derive the net model of a tool. Then, all sequences of operations leading to deadlocks are identified at the level of changes of configurations and are eliminated from further considerations. Only deadlock-free sequences are chosen to be analyzed with respect to their performance.

Lee and Lee [71] develop POPN models for single-arm cluster tools with wafer revisiting. In their models, some operations performed at the same revisiting PM are separately modeled by operation places. A place with its capacity being one is used to model the availability of a revisiting PM such that if a token is in such a place, the corresponding PM is available; otherwise it is not. Also, a place modeling a revisiting PM is both the input and output one of those modeling the operations performed at such a PM. Then, a necessary and sufficient condition is developed to avoid deadlocks.

To ease the search for an optimal schedule, Wu *et al.* [120] propose an ROPN model whose usage can eliminate the need to consider infeasible robot activities, e.g., unloading a wafer from an empty module and loading a wafer into an occupied module. Based on such a model for single-arm cluster tools in [120], a PN model for dual-arm cluster tools is also developed in [121] and [130]. These models are well used to identify an optimal schedule.

C. Scheduling and Control

Without intermediate buffers between PMs in a cluster tool, revisiting processes make the scheduling problem challenging. Such a problem is similar to a hoist scheduling problem [14]–[16], [48], [79], [82]. According to [48] and [79], such revisiting processes for hoist only happen at its first and last processing steps, but several revisits and parallel PMs for a process step are not allowed in hoist scheduling. Thus,

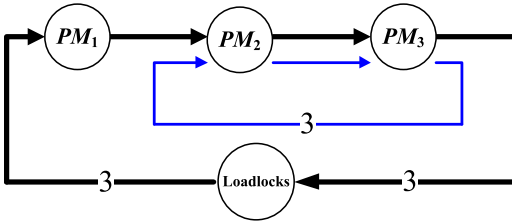


Fig. 7. Wafer flow process by using a 3-WC schedule.

hoist scheduling results are not applicable to cluster tools with revisiting processes.

Perkinston *et al.* [88] present a practical case and analyze how revisiting processes affect the system throughput and compute a rough bound on the cycle time. For the first time, the work [71] studies the scheduling problem of cluster tools with wafer revisiting and points out that the choice of the processing order, in conjunction with the robot task sequence, may cause a circular wait between a pair of chambers, each holding a wafer, where the robot tries to unload a wafer from one chamber and load it into the other. This causes a deadlock from which a cluster tool cannot proceed. Deadlock freedom is essential to operate cluster tools with revisiting processes. The backward scheduling strategy for revisiting processes may become infeasible. Hence, scheduling a single-arm cluster tool coping with revisiting processes is much more complicated than doing so without them. To find a feasible and optimal schedule, Lee and Lee [71] formulate the problem as a mixed integer program. But, it is not computationally efficient because its complexity is exponential with the number of discrete variables.

For a single-arm cluster tool with wafer flow pattern $(m_1, \dots, (m_p, m_{p+1})^h, \dots, m_n)$, $2 < p + 1 \leq n$, when $m_i < 3$, $i \in \{p, p + 1\}$, based on an ROPN model, Wu *et al.* [120] conclude that, for revisiting pattern $(1, 1)^h$, the optimal periodic schedule is unique. For $(1, 2)^h$ and $(2, 1)^h$, the optimal one can be obtained by selecting the best one among h schedules. With optimal deadlock avoidance policy, an optimal periodic schedule for the case $h = 2$ is derived by using analytical expressions. Their method is much more efficient than the mixed integer programming method [71]. The bigger h is, the more complicated the system scheduling is.

A dual-arm cluster tool can help reduce the cycle time for transport-bound cases by enabling a swap scheduling strategy. With two arms available, such a strategy is efficient. It simplifies robot task scheduling and shortens robot cycle time. Compared with the backward strategy for single-arm cluster tools, a swap strategy for dual-arm cluster tools with revisiting can still be applied without causing any deadlock problem. Zuberek [156] develops POPN models for dual-arm cluster tools coping with revisiting processes. After simple parallel paths are eliminated, the models become straightforward. Although they are useful as a scheduling guide and tool to verify the performance of a given schedule, no method is established to find an optimal schedule.

Wu *et al.* [121] find that, for dual-arm cluster tools under a swap strategy, revisiting processes contain different cycles.

If a cycle involves all processing steps of the system, it is called a global cycle. Otherwise, it is called a local cycle if it involves the revisiting processes only. The system repeatedly switches from the global cycles to local ones or from the local to global ones. Under a swap strategy, three wafers are completed in a period for an example tool, called a three-wafer cyclic (3-WC) schedule, containing three local and three global cycles. In detail, after undergoing three local cycles, the system switches to a global one. Then, after three global cycles, it switches to the local one again. The process repeats, as shown in Fig. 7. Under some conditions, the system is always in a transient state instead of a steady state. With this important finding, methods are proposed to calculate the correct cycle time in [99] and [121].

When the system switches from local to global cycles, there is a delay for unloading a processed wafer at a revisiting step, which increases the cycle time of the system. Thus, the obtained schedule via the method in [121] is not optimal. Wu *et al.* [130] reveal that decreasing the number of local and global cycles in a cyclic schedule can improve the throughput performance of revisiting processes. Then, their work proposes a two-wafer cyclic (2-WC) scheduling strategy that contains two global cycles and two local cycles. Based on it, cycle time analysis is carried out to evaluate the tool's performance. It is concluded that a 2-WC scheduling strategy can improve the productivity of the system.

More local and global cycles lead to a longer delay for unloading a processed wafer at a revisiting step. Thus, it is highly needed to find a scheduling method to derive a schedule with the fewest global and local cycles. A 1-WC schedule contains one local cycle and one global cycle only, called a 1-WC schedule, and it should be the best. To derive a 1-WC schedule and verify this conclusion, Qiao *et al.* [94] propose a strategy called modified swap-based strategy for a dual-arm cluster tool with the wafer flow pattern $(1, (1, 1)^2)$. Furthermore, based on 1-WC schedule, Qiao *et al.* [96], [98], [101] propose efficient scheduling methods for time-constrained dual-arm cluster tools with wafer revisiting or both wafer revisiting and activity time variation, while Pan *et al.* [84] develop two heuristic algorithms to optimize the start-up transient processes for dual-arm cluster tools with wafer revisiting.

V. FUTURE RESEARCH

It can be seen that, in recent years, extensive studies have been done on scheduling of cluster tools in semiconductor manufacturing and significant advancements for some challenging problems are made. However, with technology improvement and market demand changes, there are still many challenging issues. New techniques are necessary to meet the practical industry requirements as pointed out below.

A. Lot Switching Scheduling

The existing studies on cluster tool scheduling are done mainly for finding a cyclic schedule under the steady state. Recently, with larger wafer diameter and stricter fabrication requirement, more complicated wafer flow patterns need to be handled. Furthermore, smaller batches become more popular, thereby requiring cluster tools to switch more frequently

TABLE I
COMPARISON OF MODELING METHODOLOGIES WITHOUT WAFER REVISITING

Typical reference	Tool type	Modeling method	States	Advantages	Disadvantages
Perkinston <i>et al.</i> [87]	Single-arm cluster tools	Time diagram	Steady state and transient processes	Simple	Hard to describe complex systems
Venkatesh <i>et al.</i> [114]	Single-arm and dual-arm tools	Time diagram	Steady state	Simple	Hard to describe complex systems
Yim and Lee [144]	Single-arm multiple cluster tools	Using the job requirement table and the candidate list	Transient processes	Describe the routing flexibility	Hard to describe complex systems
Zuberek [155]	Single-arm and dual-arm cluster tools	POPN Model	Steady-state and transient processes	Easily and clearly describe manufacturing processes	Depend on the wafer flow pattern
Kim <i>et al.</i> [58] and Lee and Park [73]	Dual-arm cluster tools	POPN Model	Steady state	Easily and clearly describe manufacturing processes	Depend on the wafer flow pattern
Wu <i>et al.</i> [119]	Single-arm tools	ROPN model	Steady state	Compact and generic model	Complicated and abstract model
Wu and Zhou [125]	Dual-arm cluster tools	ROPN model	Steady state	Compact and generic model	Complicated and abstract model
Kim and Lee [51]	Dual-arm Cluster tools	POPN model	Real-time	Easily and clearly describe manufacturing processes	Depend on the wafer flow pattern
Wu and Zhou [126]–[128]	Dual-arm Cluster tools	ROPN model	Real-time	Compact model	Complicated and abstract model
Qiao <i>et al.</i> [92], [93]	Single-arm Cluster tools	ROPN model	Real-time	Compact model	Complicated and abstract model
Lee <i>et al.</i> [67], [68]; Kim <i>et al.</i> [53], [56]	Single-arm and dual-arm Cluster tools	POPN Model	Transient processes	Easily and clearly describe manufacturing processes	Depend on the wafer flow pattern

between a transient state and steady one. Such switches can lead to more deadlock issues. Although some important efforts have been made for this issue [49], [53]–[57], [66]–[68], [100], [115], [154], most of them involve complex computation. Finding more efficient methods is a meaningful research topic. Furthermore, with wafer residency time constraints, it is very difficult to schedule transient processes of cluster tools, especially when the activity time is subject to random disturbance. Thus, more attention should be paid to transient process scheduling for different situations.

B. Cluster Tools With Multiple Wafer Types Being Concurrently Processed

Due to process engineering considerations, restrictions in gas supply piping to tool chambers and quality risk reduction, fewer process steps for processing a wafer lot in a tool tend to be used in semiconductor manufacturing, typically two or three, for earlier wafer quality inspection even though the tool has more PMs [69], [70]. This implies that parallel PMs are configured for a process step. In order to increase utilization of PMs, multiple wafer types, which may have similar but different recipes, wafer flow patterns, and process time, are concurrently processed in a tool.

Although the existing methods for periodic scheduling can be extended to schedule cluster tools with multiple wafer types concurrently processed under the steady state [69], [70] with small batches, a cluster tool often operates in a transient process. Thus, it is important to schedule lot switching processes of cluster tools dealing with multiple wafer types. This is an open scheduling problem. In addition, engineers

from semiconductor fabs indicate that the impacts of processing time variation on cluster tools are significant. How to reduce such impacts on cluster tools as much as possible is another open problem requiring further research.

C. Dealing With Chamber Cleaning Requirements

Without frequently removing residual chemicals on the surface of chamber wall and impurities in the chamber of a PM, wafers may be damaged by the contamination of such chemicals. In order to improve the quality of wafers, it is important to clean chambers in PMs periodically [50]. The chamber cleaning is used for all CVD tools to guarantee the quality of wafers. It is a critical step in the thin film deposition process as well.

A chamber should be cleaned after a specified number of wafers are processed (i.e., periodic cleaning step) or when significant contamination within a chamber is detected. With chamber cleaning requirements, the widely applied backward strategy for single-arm cluster tools and swap strategy for dual-arm cluster tools may become non-optimal if periodical chamber cleaning is performed. Furthermore, chamber cleaning requirements together with other factors, such as wafer residency time constraints, failure-prone PMs, activity time variation, and other disruptive events, further complicate the operation of cluster tools. Thus, new strategy and methodology are necessary, which requires further research.

D. PM Failure Response Policies

With wafer residency time constraints, any failure of PMs may make the existing schedule infeasible. Thus, it is crucial to respond to a PM failure such that the system can keep

TABLE II
COMPARISON OF SCHEDULING METHODOLOGIES WITHOUT WAFER REVISITING

Typical reference	Tool type	Scheduling method	Advantages	Disadvantages	Objective
Yim and Lee [144]	Single-arm multiple cluster tools	Simulated annealing (SA)	Optimal or near optimal schedule	Inefficiency	Minimum makespan
Rostami <i>et al.</i> [106]	Dual-arm cluster tools	A search method combined with heuristics	Find a feasible schedule	Inefficiency	Maximum throughput
Lee and Park [73]	Dual-arm cluster tools	A mixed integer programming model	Optimal schedule	Inefficiency for a large scale size problem	Minimum cycle time
Yi <i>et al.</i> [142]	Multi-cluster tools	Polynomial algorithm	Optimal schedule	Without considering wafer residency time constraints and activity time variation	Maximum Steady state throughput
Zhu <i>et al.</i> [151]	Multi-cluster tools	Polynomial algorithm	Optimal one-wafer cyclic schedule	Without considering wafer residency time constraints and activity time variation	Minimum cycle time
Wu <i>et al.</i> [119]	Single-arm cluster tools	Robot waiting time method	Optimal schedule	Without considering activity time variation	Minimum cycle time
Wu and Zhou [125]	Dual-arm cluster tools	Robot waiting time method	Optimal schedule	Without considering activity time variation	Minimum cycle time
Kim and Lee [51]	Dual-arm cluster tools	Branching technique	Address the schedulability and how to find a feasible schedule	Some cases identified by the schedulability which are not schedulable are schedulable in fact	Schedulability and scheduling analysis
Wu and Zhou [126]–[128]	Dual-arm cluster tools	Real-time robot waiting time method	Optimal schedule	Schedulability conditions are sufficient	Schedulability and scheduling analysis
Qiao <i>et al.</i> [92], [93]	Single-arm cluster tools	Real-time robot waiting time method	Optimal schedule	Schedulability conditions are sufficient	Schedulability and scheduling analysis
Qiao <i>et al.</i> [95], [97]	Single-arm cluster tools	Robot waiting time method	Find a feasible transient schedule	The obtained transient schedule is not optimal	Dealing with PM failure
Lee <i>et al.</i> [67], [68]; Kim <i>et al.</i> [53]	Single-arm and dual-arm cluster tools	Mixed integer programming models	Optimal schedule	Without considering wafer residency time constraints and activity time variation	Minimum makespan

working in a feasible way. To do so, Qiao *et al.* [95], [97] propose PM failure response policies for single-arm cluster tools. Such response policies are composed of some simple control laws which can be implemented in practice. However, the scheduling strategy for single-arm cluster tools is different from the one for dual-arm cluster tools. The research results in [95] and [97] are not applicable to dual-arm cluster tools. There is no research report on presenting PM failure response policies for dual-arm cluster tools such that the system can keep working without violating wafer residency time constraints.

When a tool is required to close down such that it can be repaired after a failure, efficient methods are necessary to optimize such a close-down process. There are some existing studies on scheduling transient processes already in [49], [53]–[57], [66]–[68], [100], [115], and [154]. However, they cannot deal with PM failure. Thus, their methods are not applicable to the scheduling problem of close-down processes of cluster tools with a PM failure. Therefore, how to close down the cluster tool after a PM failure is still an open problem.

E. Multicenter Tools Scheduling

In [4], [137]–[140], [142], [143], and [151]–[153], efforts have been made in modeling and performance evaluation of

multicenter tools. However, these studies focus on steady-state scheduling and the research on the operation of multicenter tools is at the infant stage. A multicenter tool is structurally more complex than a single-cluster tool. It is much more complicated to deal with activity time disturbance, wafer residency time constraints, lot switching, transient processes, and PM failures. Up to now, there is no any research report on these issues. Thus, this is an important research direction.

There are some excellent research results in scheduling cluster tools. These results may be extended to other automated manufacturing systems, especially for the systems in which the operations of transportation facilities have effects on the system throughput.

VI. CONCLUSION

This paper reviews the existing results that address the modeling and scheduling issue of cluster tools in semiconductor manufacturing. Cluster tools consist of several process chambers and a single or dual-arm robot located in the center of a tool. Due to the configuration of the tools, it is very important to schedule the robot tasks and wafer processing activities simultaneously. For nonrevisiting processes, with some constraints and no buffers between PMs, it is difficult to schedule cluster tools. Thus, this paper surveys scheduling

and modeling methodologies for cluster tools with nonrevisiting processes. As revisiting processes share some chambers by multiple processing steps, the operations of cluster tools dealing with them are more complicated than those with nonrevisiting processes only. Deadlock and blocking problems in revisiting processes should be addressed. To satisfy wafer fabrication requirements in practice, we have to find and implement the optimal schedule for cluster tools with revisiting processes by considering complicated wafer flow patterns, real time scheduling, and residency time constraints such that the cycle time is minimized even if the activity time variations occur. This is vitally important to increase a semiconductor manufacturer's competitiveness from a long run. After the advances are reviewed, we point out some existing challenge issues and future research problems.

APPENDIX

See Tables I and II.

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